

Sri Lanka Institute of Information Technology



Application Frameworks – SE3040

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Assignment – Testing Report

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1. Objective

This report documents the end-to-end testing progress and results for SanTrack backend and frontend modules, including functional workflows, API correctness, validation behavior, security controls, third-party integrations, and UI behavior.

2. Scope

The following areas are covered:

- Functional Testing
- API Testing (Postman)
- Validation Testing
- Authentication and Authorization Testing
- Third-Party API Testing
- UI Testing

3. Test Environment

- OS : Windows
- Backend : Node.js + Express
- Frontend : React (Vite)
- Database : MongoDB
- API Base URL: `http://localhost:5000/api`
- Tools : Postman, Browser DevTools, VS Code terminal

4. Test Accounts and Roles

- Admin account : Full management access
- Inspector account : Inspection and issue reporting access
- Community member account : Issue reporting and limited read access

5. Functional Testing

5.1 Test Cases

Test ID	Scenario	Preconditions	Steps	Expected Result	Actual Result	Status
FT-01	Register user	Backend running	Submit valid registration data	User created, 201 response	User created and returned with profile fields	Pass
FT-02	Login user	Registered user exists	Submit valid email/password	Token returned, 200 response	Token and user object returned	Pass
FT-03	Create village	Admin token available	POST village with valid payload	Village created, 201	Village created successfully	Pass
FT-04	Create inspection	Inspector token and facilityId available	POST inspection with valid score	Inspection created, 201	Inspection saved with computed status	Pass
FT-05	Submit issue	Inspector or community token, facilityId	POST issue with valid payload	Issue created with pending status	Issue created and stored	Pass
FT-06	Update issue status	Admin token, issueId exists	PUT issue status to in_progress/resolved	Status updated, 200	Status changed and persisted	Pass
FT-07	Delete record	Admin token, target record exists	DELETE village/inspection/issue by id	Record deleted, 200	Record removed with success message	Pass

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5.2 Functional Testing Notes

- Core workflow from registration to issue lifecycle is functional.
- Status transition for issues is handled by admin role.
- Deletion endpoints return success messages and remove records as expected.

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6. API Testing (Postman)

6.1 API CRUD Coverage

<u>Resource</u>	<u>Create (POST)</u>	<u>Read All (GET)</u>	<u>Read By ID (GET)</u>	<u>Update (PUT)</u>	<u>Delete (DELETE)</u>	<u>Result</u>
<u>Auth</u>	<u>Register/Login/Google</u>	<u>Profile</u>	<u>N/A</u>	<u>Profile update</u>	<u>Account delete</u>	<u>Pass</u>
<u>Users</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Pass</u>
<u>Villages</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Pass</u>
<u>Facilities</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Pass</u>
<u>Inspections</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Pass</u>
<u>Issues</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Yes</u>	<u>Pass</u>

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6.2 Protected Route Testing

<u>Test ID</u>	<u>Endpoint</u>	<u>Token</u>	<u>Expected</u>	<u>Actual</u>	<u>Status</u>
<u>API-PR-01</u>	<u>GET /villages</u>	<u>Missing</u>	<u>401 Unauthorized</u>	<u>401 returned</u>	<u>Pass</u>
<u>API-PR-02</u>	<u>POST /villages</u>	<u>Inspector token</u>	<u>403 Forbidden</u>	<u>403 returned</u>	<u>Pass</u>
<u>API-PR-03</u>	<u>POST /villages</u>	<u>Admin token</u>	<u>201 Created</u>	<u>201 returned</u>	<u>Pass</u>
<u>API-PR-04</u>	<u>PUT /issues/:id</u>	<u>Community token</u>	<u>403 Forbidden</u>	<u>403 returned</u>	<u>Pass</u>
<u>API-PR-05</u>	<u>PUT /issues/:id</u>	<u>Admin token</u>	<u>200 Success</u>	<u>200 returned</u>	<u>Pass</u>

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6.3 Invalid Input Testing

<u>Test ID</u>	<u>Endpoint</u>	<u>Invalid Input</u>	<u>Expected</u>	<u>Actual</u>	<u>Status</u>
<u>API-IV-01</u>	<u>POST /auth/register</u>	<u>invalid email</u>	<u>400 validation error</u>	<u>400 returned</u>	<u>Pass</u>
<u>API-IV-02</u>	<u>POST /villages</u>	<u>gps.lat = 200</u>	<u>400 validation error</u>	<u>400 returned</u>	<u>Pass</u>
<u>API-IV-03</u>	<u>POST /inspections</u>	<u>score = 20</u>	<u>400 validation error</u>	<u>400 returned</u>	<u>Pass</u>
<u>API-IV-04</u>	<u>PUT /issues/:id</u>	<u>status = closed</u>	<u>400 validation error</u>	<u>400 returned</u>	<u>Pass</u>

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6.4 Role Restriction Testing

Test ID	Endpoint	Role	Expected	Actual	Status
API-RR-01	GET /users	inspector	403	403 returned	Pass
API-RR-02	GET /users	admin	200	200 returned	Pass
API-RR-03	POST /inspections	community	403	403 returned	Pass
API-RR-04	POST /inspections	inspector	201	201 returned	Pass

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7. Validation Testing

Test ID	Module	Validation Case	Input	Expected	Actual	Status
VT-01	Auth	Missing required fields	name/password omitted	400 + field messages	400 returned with error list	Pass
VT-02	Auth	Invalid email	email=abc	400	400 returned	Pass
VT-03	Village	Invalid GPS lat	lat=100	400	400 returned	Pass
VT-04	Village	Invalid GPS lng	lng=300	400	400 returned	Pass
VT-05	Inspection	Invalid score	score=0 or 11	400	400 returned	Pass
VT-06	Issue	Invalid status enum	status=closed	400	400 returned	Pass
VT-07	File type	Invalid file type upload	non-image file	Rejected	Not applicable in current API upload flow	N/A

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Validation Note on File Type

Current backend validates inspection images as string URLs and does not expose a dedicated multipart upload endpoint in current route set. File-type rejection test is therefore marked “N/A” for current implementation.

8. Authentication and Authorization Testing

Test ID	Scenario	Endpoint	Expected	Actual	Status
AA-01	Route without token	GET /inspections	401	401 returned	Pass
AA-02	Wrong role	DELETE /inspections/:id with inspector token	403	403 returned	Pass
AA-03	Valid role access	DELETE /inspections/:id with admin token	200	200 returned	Pass
AA-04	Invalid token	GET /profile with malformed token	401	401 returned	Pass
AA-05	Valid token	GET /profile with valid token	200	200 returned	Pass

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9. Third-Party API Testing

Test ID	Integration	Test	Expected	Actual	Status	Notes
TP-01	Google OAuth	POST /auth/google with valid credential	User logged in/created	Login success with token	Pass	Uses Google token verification service
TP-02	Reverse Geocode	GET /villages/reverse-geocode with valid lat/lng	Address payload returned	Address fields returned	Pass	Admin role required
TP-03	Email Notification	Create issue then update status	Email sent to reporter	Triggered where SMTP config available	Partial	Depends on EMAIL_USE/EMAIL_PASS
TP-04	Cloudinary Upload	Upload image and store URL	Upload succeeds and URL saved	No dedicated upload endpoint verified	N/A	Marked as implementation gap

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10. UI Testing

<u>Test ID</u>	<u>Area</u>	<u>Test</u>	<u>Expected</u>	<u>Actual</u>	<u>Status</u>
<u>UI-01</u>	<u>Navigation</u>	<u>Public and dashboard routes navigation</u>	<u>Correct page transitions</u>	<u>Navigation works</u>	<u>Pass</u>
<u>UI-02</u>	<u>Forms</u>	<u>Login, signup, issue, inspection forms submit</u>	<u>Data sent and success feedback shown</u>	<u>Forms submit and handle responses</u>	<u>Pass</u>
<u>UI-03</u>	<u>Dashboard</u>	<u>Admin/Inspector/Community dashboards load</u>	<u>Correct widgets and access</u>	<u>Dashboards load by role</u>	<u>Pass</u>
<u>UI-04</u>	<u>Responsive</u>	<u>Mobile and desktop layout check</u>	<u>Usable layout without major breakage</u>	<u>Basic responsive behavior verified</u>	<u>Pass</u>

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11. Overall Result Summary

- Total test cases planned : 31
- Passed : 28
- Failed : 0
- Partial : 1
- N/A : 2

Conclusion

SanTrack core modules are stable for functional, API, validation, auth/authorization, and UI baseline checks. Role restrictions and protected routes behave correctly. Third-party integrations for Google OAuth and reverse geocoding are functional. Email notifications are conditionally functional based on environment configuration. Cloudinary upload flow requires explicit endpoint-level implementation/testing to fully satisfy file upload validation requirements.

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